

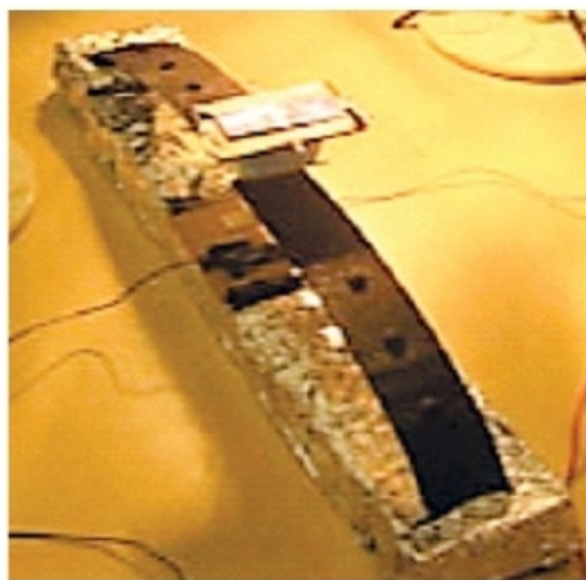


Fig. 8: Arch-type heliotropic solar cell

developed by several research institutes. In one system, six panels made from liquid crystalline elastomer for mechanical actuation, single wall carbon nanotubes for photo-thermal energy conversion and a polyurethane mesh for structural support are

assembled together to make the system. The elastomer, when heated above 68°C, changes its nematic order, leading to reversible axial contraction and mechanical actuation, enabling the system to be heliotropic.

Six composite structures made from these photo-thermal smart materials that concentrate light and act as an heat collector are bonded to six actuator beams that actuate (shrink) when incident sunlight increases temperature to about 68°C.

The above multi-stage design divides actuators into multiple units, each covering a certain angle. As the Sun moves during the day, one or two actuators that are facing the sun respond and pull the solar panel towards the Sun. These composite structures support a solar panel and move to track the Sun. This arrangement is shown in Fig. 7.

In another system, developed by Massachusetts Institute of Technology, solar panels are installed on top of a curved arch made of two kinds of metal, namely aluminium and steel. The mechanism relies on difference in coefficients of thermal expansion of the two metals. Expansion is caused by the temperature gradient between sunny and shaded areas. The light warming up one side causes the arch to bend, tilting the panels towards the light. This system has been reported to be maintenance-free. The arrangement is shown in Fig. 8.

Estimated cost

Presently, the monocrystalline silicon solar system with the ability to track the Sun, including storage battery, electrics and control system in line with the design shown in Fig. 3, is sold at about US\$ 10/W (about ₹ 720/W), whereas a conventional monocrystalline solar system without tracking device but with storage battery and electrics costs about ₹ 70,000/kW, that is, ₹ 70/W. It is estimated that cost of SunBOT solar panels with electrics, when mass manufactured, will be compatible with conventional units. This is because there is no need to provide costly GPS tracking mechanism—instead, cost-effective smart polymers would be used. Further, it would change the scenario of green power generation as these smart devices would produce about forty per cent more electricity compared to conventional solar roof installations. **EFY**

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